

The Wind-AE windsoln (seed) file format

A converged Wind-AE wind solution is stored as a single CSV-like file: a block of comment-prefixed header lines that carry every parameter, boundary condition, and the stellar spectrum, followed by the solution profile itself as a table of normalized (code-unit) columns. The same format is used by (i) the original Wind-AE Python wrapper (Murray-Clay et al. 2009; Broome et al. 2025), (ii) the bundled solution grid under `inputdata/windae_grid/`, and (iii) EXHALE's Fortran port, whose `load_seed` reads such a file as the starting point of a continuation and whose `dump_seed` writes one back. This note is the authoritative field-by-field reference, taken from the Python reader (`wrapper/wrapper_utils/windsoln.py`).

1. File layout

A windsoln file has three kinds of lines, in order:

1. **Header lines** `#key: v1,v2,...` — one per key, carrying scalars, parameters, BCs, flags, and spectrum meta-data (Table 1).
2. **Spectrum rows** `## e,wPhi, $\sigma_1$ , $\sigma_2$` — npts of them, holding the stellar spectrum sampled at the Gauss–Legendre quadrature points (§3.).
3. **Data rows** — the solution profile, one grid point per line, columns named by `#vars` and normalized by `#scales` (§4.).

Any other line beginning with `#` is a free comment and is skipped. A minimal excerpt:

```
#nspecies: 2
#vars: r,rho,v,T,Ys_HI,Ys_HeI,Ncol_HI,Ncol_HeI,q,z
#scales: 8.34e9,1.0e-15,9.08e5,1.0e4,1,1,1.0e17,1.0e17,1,8.34e9
#plnt_prms: 2.348e30,8.34e9,1.870e33,4.937e11,2.42e4,3.828e33
#phys_prms: 0.800,0.200,HI,HeI,1.673e-24,6.646e-24,2.30
#bcs: 1.030,7.755,2.466e4,0.198,1.0,1.0,1.08e-3,2.05e-4,1.85e-5,1.50e-5
#tech: 1.0,1.0e2,2.5,0.99
#flags: 1,1.00000,0.00000,1
#add_prms: 0
#spec_prms: 59,2,2009-01-01,scaled-solar,full,5.99e-8,9.12e-6,...,2.18e-11,3.94e-11
## 2.150e-9,1.771e6,4.258e-24,1.267e-22
## ... (59 rows) ...
1.030,2.466e4,2.232e-6,0.198,1.0,1.0,7.34e3,4.70e2,0.0,2.76
... (one row per grid point) ...
```

Table 1: Header lines of a windsoln file. Field order is exact; “ $\times N$ ” marks a block repeated for each species.

Key	Fields	Meaning (and units)
#nspecies	N	number of species ($N = 2$: H I, He I).
#vars	names	data-grid column labels (Table 2).
#scales	N_{col}	cgs value of one code unit for each data column.
#plnt_prms	6	$M_p[\text{g}]$, $R_{\text{pl}}[\text{cm}]$, $M_\star[\text{g}]$, $a[\text{cm}]$, $F_{\text{tot}}[\text{erg cm}^{-2}\text{s}^{-1}]$, $L_\star[\text{erg s}^{-1}]$. <i>Field 5 is the total ionizing flux, not T_{eq}.</i>
#phys_prms	$3N+1$	mass fractions X_s ($\times N$); species names ($\times N$); atomic masses [g] ($\times N$); then molec_adjust (mean-molecular-weight boost of the molecular base layer, ≈ 2.3).
#bcs	$2N+6$	$R_{\text{min}}, R_{\text{max}}$ [R_{pl}]; ρ_{min} [ρ_0]; T_{min} [T_0]; base neutral fractions $Y_{s,\text{min}}$ ($\times N$); sonic-point columns $N_{\text{col},s}$ [$N_{\text{col},0}$] ($\times N$); erf_drop (2: the erf-velocity molecular $\rightarrow$ atomic transition).
#tech	4	breezeparam, rapidity, erfn, mach_limit (relaxation/solver controls).
#flags	4	lyacool (0/1); tidalforce (0 = spherical, 1 = tidal/Roche); bolo_heat_cool (0 = atomic base, 1 = molecular bolometric layer); integrate_outward (0/1).
#add_prms	$1+2n$	n , then n (name,value) pairs of optional extra parameters ($n = 0$: none).
#spec_prms	meta	npts, N , date, source file (scaled-solar), kind (full), window[2], resolved[2], normalized[2] (wavelength bounds [cm]), ionization potentials χ_s [erg] ($\times N$). See §3..

2. Header reference

Table 1 lists every header key. N is the number of species (#nspecies; $N = 2$ for the bundled H/He data). Planet parameters are in cgs; boundary conditions and the data grid are in the code units of Table 2.

3. The embedded stellar spectrum

The stellar XUV spectrum that the solution was computed with is stored in the file itself, in two parts:

- #spec_prms — metadata: the number of quadrature points npts, the source label (here scaled-solar, a solar-like SED), the spectral kind (full, i.e. resolved rather than monochromatic), the integration window in wavelength ($\approx [6, 912]\text{\AA}$, i.e. X-ray to the H Lyman limit), and the ionization potentials χ_s for each species (e.g. $2.18 \times 10^{-11}\text{ erg} = 13.6\text{ eV}$ for H, $3.94 \times 10^{-11}\text{ erg} = 24.6\text{ eV}$ for He).
- the ## rows — the spectrum proper, one row per quadrature point: photon energy hc/λ_i [erg]; the *normalized* weight $w_i \Phi_{\lambda_i}/F_{\text{tot}}$; and the photoionization cross sections $\sigma_{\lambda_i,s}$ [cm^2] for each species.

Table 2: Data-grid columns (#vars) and their code→cgs scales (#scales). $\rho_0=10^{-15}$, $c_{s,0}=\sqrt{k_B T_0/m_H}=9.08 \times 10^5$, $T_0=10^4$, $N_{\text{col},0}=10^{17}$.

Column	Symbol	Scale	Meaning
r	r	R_{pl}	radius
rho	ρ	ρ_0	mass density [g cm^{-3}]
v	v	$c_{s,0}$	radial velocity [cm s^{-1}]
T	T	T_0	temperature [K]
Ys_HI	Y_{HI}	1	H neutral fraction
Ys_HeI	Y_{HeI}	1	He neutral fraction
Ncol_HI	N_{HI}	$N_{\text{col},0}$	H column to the star [cm^{-2}]
Ncol_HeI	N_{HeI}	$N_{\text{col},0}$	He column to the star [cm^{-2}]
q	q	1	relaxation coordinate $\in [0, 1]$
z	z	R_{pl}	base→sonic span ($r = R_{\text{min}} + qz$)

The crucial point is the separation of *shape* and *amplitude*: the ## weights are normalized by F_{tot} , so they encode only the spectral *shape* (which photon energies carry what fraction of the ionizing flux), while the absolute level is set by the single scalar F_{tot} in #plnt_prms. This is why the bundled grid can keep one fixed scaled-solar shape across all ~ 1000 solutions and vary only F_{tot} (and M_p, R_{pl}) from planet to planet. Verified: #spec_prms is byte-identical across the entire grid.

Caveat (spectrum dependence). The wind structure depends on the SED shape, not only on F_{tot} . The bundled data assume a scaled-solar spectrum; an EXHALE run that instead assumes, say, a power law produces different heating/ionization profiles. A Wind-AE warm-start is therefore only self-consistent with an EXHALE run that uses the *same* spectrum. Conversely, a power-law EXHALE model failing to converge does *not* imply that no steady wind exists for that planet — only that that particular spectrum/numerics combination did not settle.

4. The solution grid

Each data row is one grid point; the columns are named by #vars and are stored in dimensionless code units. The physical (cgs) value is recovered as (physical) = (code) $\times$ (scale), with the scale taken from #scales (Table 2). The independent variable is the relaxation coordinate $q \in [0, 1]$ mapping the base to the sonic point; the physical radius is $r = R_{\text{min}} + qz$, where z (constant along the solution) is the base→sonic span in R_{pl} .

5. Use in the EXHALE port

src/modules/wind_ae/wae_continuation.f90 reads and writes this format:

- load_seed parses #plnt_prms, #phys_prms, #bcs, #tech, #flags, and the first m data rows (into the relaxation arrays). It *ignores* #scales, #add_prms, #spec_prms, and the ## rows: the spectrum is loaded separately from

inputdata/windae_spectrum.inp, which is the same scaled-solar data in the equivalent “# KEY: . . .” header form.

- **dump_seed** is the exact inverse: it reconstructs the full header (including #scales, #spec_prms, and the ## block from the in-memory spectrum arrays) and writes the m relaxation rows, so a converged solution can be banked as a reusable seed.
- **Seed library and picker.** inputdata/windae_grid/ holds the Broome et al. (2025) grid (~ 1000 solutions spanning $M_p \in [3, 528] M_\oplus$ and $R_{pl} \in [1.9, 22.6] R_\oplus$, all with $\text{tidalforce}=1$ and the fixed scaled-solar spectrum), plus the eight starting_solutions. With Wind-AE seed: grid in input.inp, pick_nearest_seed reads the grid manifest (one row of path, M_p , R_{pl} , F_{tot} per file; rebuilt by make_windae_manifest.sh) and selects the solution closest in $(\log M_p, \log R_{pl})$, so the continuation starts from a near neighbor rather than the single distant fiducial seed. Wind-AE seed out: dumps the converged solution back into this format.